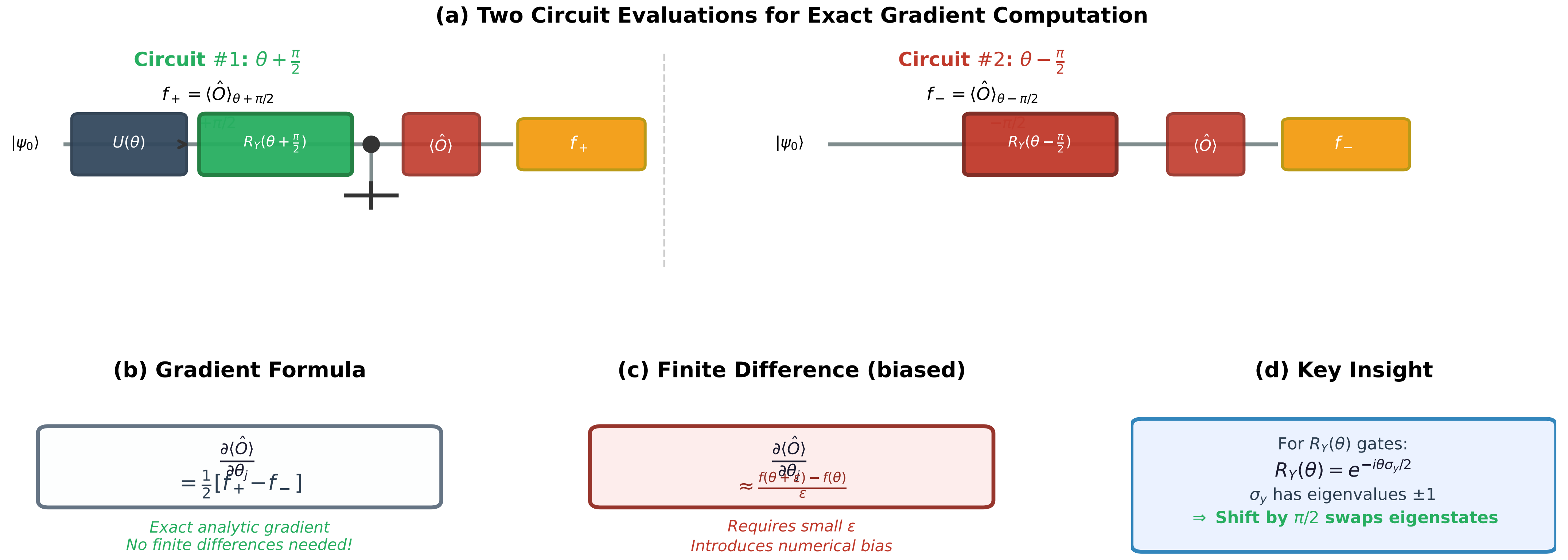

Fig QC-3: Parameter-Shift Rule gradient computation. The exact analytic gradient $\partial\langle\hat{O}\rangle/\partial\theta$ is obtained from two circuit evaluations with shifted angles $\theta \pm \pi/2$, requiring no finite-difference approximation. The $\frac{1}{2}$ prefactor arises from the σ_y generator eigenvalues.